

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Sixth Semester of B. Tech IT Examination
University Theory Examination (Supplementary Exam - 1) June 2022
IT355 Software Engineering

Date: 25.06.2022, Saturday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of a scientific calculator is allowed.

SECTION - I**Q - 1 Do as directed.**

- (i) Which one of the following is NOT desired in a good Software Requirement Specifications (SRS) document? [07]
 (A) Functional Requirements (B) Non-functional Requirements
 (C) Algorithms for Software Implementation (D) Goals of Implementation
- (ii) Which of the following are not considered when computing function point for a software project? [01]
 (1) External input and output
 (2) Programming language
 (3) User Interface
 (4) External Interface
 (5) No of programmers
 (6) files used
 (A) 02,03 (B) 01,05 (C) 04,06 (D) 02,05
- (iii) In agile process models requirements engineering and design activities are interleaved. True or False? Give Reason. [01]
- (iv) In a software project, COCOMO (Constructive Cost Model) is used to estimate [01]
 (A) Effort and duration based on size of software.
 (B) Size and duration based on effort of software.
 (C) Effort and cost based on duration of software.
 (D) Size, effort and duration based on cost of software.
- (v) Explain which process model is most suitable for the following definition and justify it. [03]
 1. A new library automation software that would link various libraries in the city.
 2. E-governance System.
 3. Sudoku Game.

Q-2.a List different process models and explain the iterative waterfall model in detail. [04]

OR

Q-2.a Explain the COCOMO model with an example. [04]

Q-2.b Answer any TWO questions.

- (i) Explain Spiral Model with a neat sketch and its advantages over other process models in detail. [10]
 - (ii) What are the umbrella activities of the software process? Explain any two in detail.
 - (iii) List out different types of software architectural styles. Discuss any two in detail.
- Q-3.a** What are the factors affecting software crises? List characteristics of a good SRS document. [04]

OR

Q-3.a List various types of risk and explain any two with examples.

Q-3.b Answer any TWO questions.

- (i) Illustrate Boehm's WSHH Principle based on an online shopping website. [04]
- (ii) What is the management spectrum? Describe the role of the four P's in software project management. [10]
- (iii) A system being developed has the following characteristics:

Number of user inputs(I)	30
Number of user outputs(O)	60
Number of user inquiries(E)	23
Number of files(F)	8
Number of external interfaces(N)	2

It is given that the complexity weighting factors for I, O, E, F and N are 4, 5, 4, 10 and 7, respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors are not applicable, each of the other four factors have value 3, and each of the remaining factors have value 4. Calculate FP (function point) for given system.

SECTION - II

Q - 4 Do as directed.

- (i) In the context of modular software design, which one of the following combinations is desirable? [07]
 (A) High cohesion and high coupling
 (B) High cohesion and low coupling
 (C) Low cohesion and high coupling
 (D) Low cohesion and low coupling
- (ii) The context diagram is also known as [01]
 (A) Level-0 DFD
 (B) Level-1 DFD
 (C) Level-3 DFD
 (D) None of the above

- (iii) CASE stands for [01]
 (A) Cost Aided Software Engineering
 (B) Computer Aided Software Engineering
 (C) Control Aided Software Engineering
 (D) None of the above

- (iv) What is the normal order of activities in which traditional software testing is organized? [01]

- (a) integration testing (b) system testing (c) unit testing (d) Validating testing
 (A) a,d,c,b (B) b,d,a,c
 (C) c,a,d,b (D) d,b,c,a

- (v) Define the following terms.(Any 3) [03]
 1. Verification
 2. Validation
 3. Alpha Testing
 4. Beta Testing

- Q-5.a Draw UML diagrams (Use Case, Activity and Sequence) for railway reservation system [07]
 (Eg. IRCTC)

- Q-5.b i. Justify why white box testing is also called structured testing. Explain statement coverage with example. [04]
 ii. List types of maintenance and explain adaptive maintenance in detail. [03]

OR

- Q-5.b i. What is cohesion? Discuss any two types of cohesion in detail. [04]
 ii. Compare ISO 9001, 9002 and 9003 with proper example. [03]

- Q-6.a Draw context and data flow diagram (DFD) up to level 1 for Online Shopping Website [04]
 (Eg. Amazon)

- Q-6.b Answer any TWO questions. [10]
 (i) What is black box testing? Explain equivalence class partitioning technique of black box testing with example.
 (ii) Draw neat sketch of CMM levels and mention different process areas of CMM.
 (iii) Define Software configuration Management and list its activities and explain any two.
